
Pre-training a Small Korean Mini-BERT: An Ablation Study on Vocabulary Size, Model Capacity, and Masking Probability

Kim Sion¹

Abstract

Pre-trained language models such as BERT learn contextual representations from large amounts of unlabeled text, but their training requires substantial computational resources. In this work, we implement Mini-BERT, a small Korean BERT with about 1.01M parameters, and pre-train it on Korean Wikipedia with MLM and NSP objectives. After 10 epochs of training, validation loss decreased from 7.6156 to 5.0036, MLM accuracy increased from 4.1% to 23.9%, and NSP accuracy increased from 72.9% to 77.4%. We also conducted an ablation study on vocabulary size, model capacity, and masking probability. Under short training, increasing model capacity lowered validation loss and raised MLM accuracy, with the largest improvement observed when the number of encoder layers was increased from Base to Large. In contrast, the vocabulary size experiment made it difficult to isolate the effect because the parameter count and the basis for comparing loss changed together, and the masking probability experiment showed no clear difference because it was conducted before the models had been sufficiently trained.

1. Introduction

BERT (Devlin et al., 2019) is a representative method for pre-training contextual language representations on large-scale text using a Transformer encoder. BERT predicts masked tokens through Masked Language Modeling (MLM) and learns the relationship between two sentences through Next Sentence Prediction (NSP). BERT-family models have since been released for many languages, and Korean-specific pre-trained models are also available (Park et al., 2021).

However, BERT-Base has about 110 million parameters, and experiments that pre-train such a model from scratch

while varying its configuration are costly in small-scale research settings. To reduce this cost, prior work has proposed small BERT models with fewer layers and smaller hidden dimensions (Turc et al., 2019), knowledge distillation that transfers knowledge from a large model to a small one (Sanh et al., 2019), and parameter sharing with embedding factorization (Lan et al., 2020). However, these studies mainly target English or assume an already-trained large model. The motivation of this study is to understand how basic settings such as vocabulary size, model size, and masking ratio affect training when an ultra-small Korean model of about 1M parameters is pre-trained from scratch.

Accordingly, we implement a Korean Mini-BERT with about 1.01M parameters, pre-train it on Korean Wikipedia, and design our experiments around the following four research questions (RQs).

- RQ1** Is optimization of the MLM and NSP objectives observed in a Korean Mini-BERT with about 1M parameters?
- RQ2** Does increasing model capacity improve pre-training validation loss and MLM accuracy?
- RQ3** How does vocabulary size affect the early stage of training?
- RQ4** Does changing the masking probability produce a performance difference under limited training conditions?

Our results show that optimization of the MLM and NSP objectives is observed even in a 1M-scale model, and that under short training, increasing model capacity lowers validation loss and raises MLM accuracy. The largest improvement was observed when the number of encoder layers was increased from Base to Large. In contrast, the vocabulary size and masking probability experiments did not support clear conclusions due to limitations in the experimental design; this paper analyzes the causes and suggests directions for improvement.

¹AIFEL, Korea. Correspondence to: Kim Sion <sion200248@gmail.com>.

2. Related Work

2.1. BERT and Pre-training Objectives

Devlin et al. (2019) proposed BERT, which pre-trains bidirectional contextual representations with MLM and NSP. ALBERT (Lan et al., 2020) later pointed out that NSP can be solved easily from topic differences alone rather than sentence order, and proposed sentence order prediction (SOP), which determines the order of two sentences. This study uses MLM and NSP together, but, as described in Section 3.2, constructs negative pairs by swapping sentence order, which makes the task close to SOP.

2.2. Small BERT Models

Approaches to reducing the size of BERT include pre-training a small model with fewer layers and smaller hidden dimensions from scratch (Turc et al., 2019), distilling knowledge from a large model into a small one (Sanh et al., 2019), and methods such as ALBERT (Lan et al., 2020) that factorize the embedding matrix and share parameters across layers. Our Mini-BERT, at about 1M parameters, belongs to the first approach: pre-training from scratch without distillation.

2.3. Subword Tokenization and Masking Ratio

SentencePiece (Kudo & Richardson, 2018) is a language-independent tokenizer that learns a subword vocabulary directly from raw text without pre-tokenization, making it easy to apply to Korean, which has rich morphological variation. Meanwhile, BERT (Devlin et al., 2019) masks 15% of input tokens, but it has not been verified whether this ratio is also suitable for an ultra-small model of about 1M parameters. This study examines how these settings affect an ultra-small Korean model.

3. Methodology

3.1. Model Architecture

Mini-BERT follows the BERT architecture, as shown in Figure 1. Token, position, and segment embeddings are computed from the input token IDs and segment IDs and summed, then passed through L encoder layers, each consisting of multi-head self-attention and a feed-forward network. The default configuration is given in Table 1.

The target vocabulary size is 8,000, and the actual vocabulary contains 8,007 tokens including special tokens. The token embedding shares its weights with the MLM output layer. Therefore, most of the parameters added when the vocabulary grows come from the embedding matrix, which accounts for about 75.8% of all parameters (768,672) in the default model.

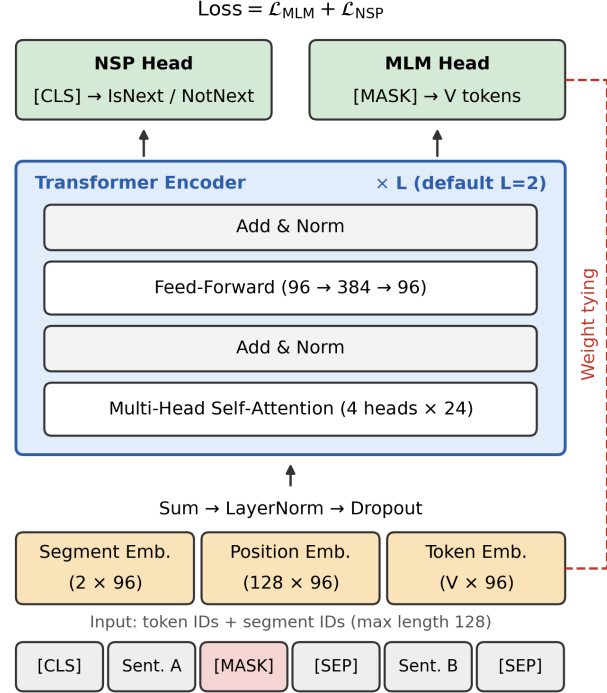


Figure 1. Mini-BERT architecture and pre-training objectives.

Table 1. Default Mini-BERT configuration.

Component	Setting	Component	Setting
Vocabulary	8,007	Batch size	64
Max length	128	Epochs	10
Hidden dim.	96	Max lr	2.5×10^{-4}
Heads (dim.)	4 (24)	Warmup	0.1
FFN dim.	384	Masking	15%
Layers	2	Dropout	0.1
Parameters	1,014,528	Optimizer	Adam

3.2. Pre-training Objectives

MLM uses whole-word masking, in which subwords grouped by whitespace are selected as a single unit, and masks up to a cap of 15% of the tokens excluding special tokens. If adding a unit would exceed the cap, that unit is skipped, so the actual masking ratio is at most 15%. Selected units are replaced with [MASK] with 80% probability, kept as the original token with 10% probability, and replaced with a random token with 10% probability. Masking is applied only once during preprocessing, so the same masked positions are used in every epoch (static masking).

NSP (sentence-order variant) splits two consecutive segments from a document into A and B, and with 50% probability uses the pair with the two segments swapped as a negative (NotNext) example. In other words, unlike the original BERT, which uses sentences from other documents as negatives, the NSP in our implementation is closer to a

Table 2. Ablation experiment conditions.

Experiment	Conditions compared	Data	Epoch
Default model	—	Full	10
Model capacity	Small / Base / Large	Full	3
Vocabulary size	4K / 8K / 16K	200K subset	3
Masking prob.	10% / 15% / 20%	200K subset	3

sentence-order discrimination task and is similar to the sentence order prediction of ALBERT (Lan et al., 2020). The [CLS] representation passes through a dense-tanh layer before being fed into a binary classifier.

3.3. Training Setup

We used Adam for optimization. The learning rate was increased to its maximum over the first 10% of all steps and then decayed to 0 with a cosine schedule. All experiments used the same random seed (1234) and were implemented in PyTorch and trained on a GPU.

4. Experimental Design

4.1. Dataset

The pre-training data was built from the Korean Wikipedia corpus, and 5% of the generated instances were randomly held out for validation. Because the split was performed at the instance level rather than the document level, instances generated from the same document may appear in both the training and validation sets. The full dataset consists of 872,265 training instances and 45,908 validation instances. Some experiments used a subset consisting of only the first 200K lines of the corpus; the number of instances generated from this subset ranges from 58,182 to 65,070 (train and validation combined), depending on the tokenizer.

4.2. Ablation Settings

We conducted three ablations, as shown in Table 2. Because of limited computational resources, training conditions differ across experiments, so performance comparisons are made only within the same experiment group, and numbers across groups are cited only as a reference for diagnosing training status. In the vocabulary size experiment, a new SentencePiece (BPE) tokenizer of each size was trained on the subset, whereas the masking experiment used the default tokenizer trained on the full corpus.

4.3. Evaluation Metrics and Comparison Baselines

We measured validation loss, MLM accuracy, and NSP accuracy. The random-guess baseline for NSP is 50%. However, when the vocabulary size differs, the number of classes to predict and the tokenization result also differ, so loss values

Table 3. Pre-training results of the default model.

Metric	1 Epoch	10 Epoch
Training loss	7.9967	5.1290
Validation loss	7.6156	5.0036
Val. MLM loss	7.1805	4.6512
Val. MLM acc.	4.1%	23.9%
Val. NSP loss	0.4350	0.3524
Val. NSP acc.	72.9%	77.4%

Table 4. Model capacity comparison (full data, 3 epochs).

Model	Hidden	Layers	Params	Val. loss	MLM acc.
Small	64	2	0.63M	6.6871	11.9%
Base	96	2	1.01M	6.3700	14.1%
Large	96	4	1.24M	5.6825	17.6%

cannot be compared directly. To account for this, we also report the MLM loss of a uniform-distribution prediction ($\ln |V|$).

5. Results

5.1. Default Model

As shown in Table 3, MLM accuracy increased substantially from 4.1% to 23.9%, whereas NSP accuracy already far exceeded the random baseline (50%) at 72.9% after 1 epoch and increased by only 4.5 percentage points thereafter. In other words, accuracy gains after the first epoch came mainly from MLM accuracy. As shown in Figure 2, validation loss decreased by 0.21 from epoch 3 to 4 but by only 0.035 in total from epoch 7 to 10, indicating that the rate of decrease slowed considerably in the later stage.

5.2. Model Capacity

Small uses hidden size 64 with 2 layers, Base uses hidden size 96 with 2 layers, and Large uses hidden size 96 with 4 layers. That is, Small→Base increases width, while Base→Large increases depth.

As shown in Table 4, loss decreased and MLM accuracy increased as the model grew. Small→Base added 0.39M parameters and reduced loss by 0.32, whereas Base→Large added 0.22M parameters and reduced loss by 0.69, yielding a larger improvement with a smaller increase in parameters.

5.3. Vocabulary Size

As shown in Table 5, validation MLM loss increased monotonically with vocabulary size, and the increase (2.03 from 4K to 16K) was larger than the increase in $\ln |V|$ (1.39). In contrast, MLM accuracy was highest for 4K, and 16K was higher than 8K, so the rankings by loss and by accuracy did

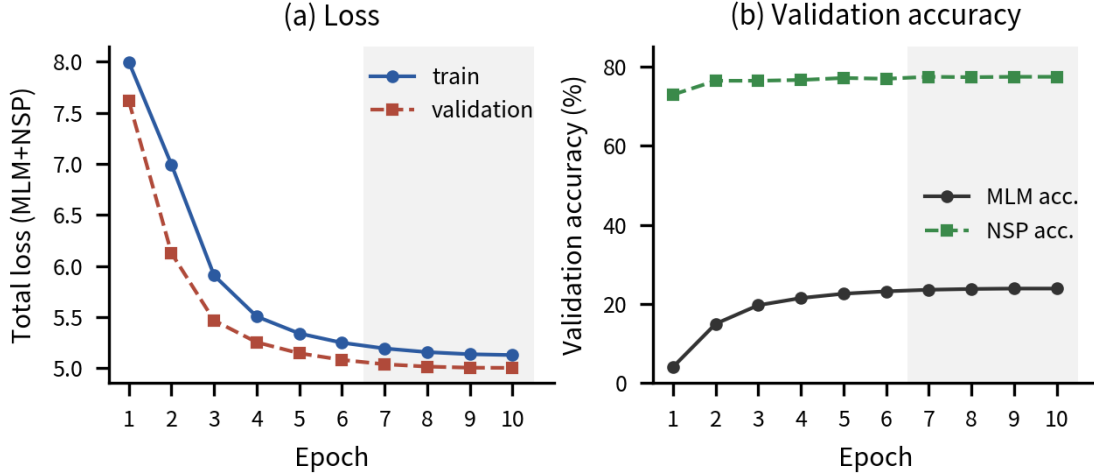


Figure 2. Per-epoch learning curves of the default model. (a) Training and validation loss (MLM + NSP). (b) Validation MLM and NSP accuracy.

Table 5. Vocabulary size comparison (200K subset, 3 epochs).

Vocab	Params	$\ln V $	MLM loss	MLM acc.
4K	0.63M	8.30	6.1668	6.3%
8K	1.01M	8.99	7.5565	2.9%
16K	1.78M	9.68	8.1958	3.7%

Table 6. Masking probability comparison (200K subset, 3 epochs).

Masking	Val. loss	MLM loss	MLM acc.
10%	8.0151	7.4144	2.8%
15%	7.9923	7.3920	2.8%
20%	7.9909	7.3896	2.8%

not agree.

5.4. Masking Probability

As shown in Table 6, the difference in validation loss across the three conditions was at most 0.0242 (at most 0.0248 for MLM loss), and MLM accuracy was 2.8% in all cases. Meanwhile, the 8K condition in Table 5 and the 15% condition in Table 6 used the same subset, vocabulary size, masking ratio, and number of epochs, but because their tokenizers were trained on different corpora (the subset vs. the full corpus), the number of generated instances (60,690 vs. 61,206) and the validation MLM loss (7.5565 vs. 7.3920) differed.

6. Discussion

6.1. Feasibility of Pre-training a 1M Model (RQ1)

The default model far exceeded the random baseline (50%) on NSP, and its MLM accuracy rose steadily from 4.1% to 23.9%. We can therefore say that training progress, in which the losses of both objectives decrease and accuracy improves, was observed even at the 1M scale. This does not mean that the model reached an optimum. Moreover, because the cosine schedule drives the learning rate close to 0 late in training, the current experiments cannot determine whether the slowdown in epochs 7–10 is due to limited model capacity or to the decaying learning rate. In addition, static masking means the model predicts the same masked positions every epoch, so a reduction in new learning signal in the later stage may also have contributed. The fact that larger models reached lower loss over the same period in Table 4 is consistent with a capacity limit, but since that is a 3-epoch comparison and even Large at 3 epochs (5.68) has higher loss than Base at 10 epochs (5.00), we consider this only indirect evidence.

6.2. Model Capacity (RQ2)

Embeddings account for about 76% of the default model’s parameters. Of the 0.39M parameters added from Small to Base, about 0.26M came from the embeddings, whereas all 0.22M parameters added from Base to Large went into encoder layers, roughly doubling the encoder parameters. In other words, under our experimental conditions, increasing encoder layers was associated with larger improvements in early pre-training metrics than increasing embeddings. However, since the hidden, head, and FFN dimensions changed together in Small→Base, the comparison does not indepen-

dently control width and depth, and each condition was a single run; this therefore cannot be generalized as an effect of encoder depth and remains a hypothesis-level interpretation.

6.3. Vocabulary Size (RQ3)

This experiment has two confounding factors. First, as the vocabulary grows, the embedding parameters grow with it. Second, the tokenization and the number of classes change, which alters the very meaning of loss and accuracy. With a small vocabulary, a single word may be split into more subwords, which can change how token-level MLM accuracy should be interpreted and may partly explain the high accuracy of 4K. In addition, the fact that loss increased more than $\ln |V|$ suggests that with a large vocabulary each token appears less often during training and may not have been learned sufficiently within 3 epochs. Furthermore, MLM accuracy barely changed over epochs 1–3 in all three conditions, so, as in Section 6.4, it must be kept in mind that this is a comparison at an early stage of training. Therefore, this result cannot be interpreted as “a smaller vocabulary is better.”

6.4. Masking Probability (RQ4)

MLM accuracy (2.8%) in all three conditions did not change over epochs 1–3, and NSP accuracy, at about 58–60%, was far below that of the full-data experiments (about 76%). The subset experiments had about 900 steps per epoch, roughly 15 times fewer updates than the full data (about 13,600 steps). Together with the fact that MLM accuracy did not change over 3 epochs, this suggests that all three conditions may have been at an early stage before sufficient training had taken place. In addition, the MLM loss difference (0.16) between the 8K condition and the 15% condition, whose generated data differ because their tokenizers were trained on different corpora, is larger than the MLM loss differences across masking conditions (within 0.03). Therefore, this result should be interpreted not as evidence that the masking ratio is unimportant, but as meaning that a difference could not be detected under the current conditions.

7. Limitations

First, as stated in Section 4.2, the data, number of epochs, and tokenizer differ across ablations, so numbers cannot be compared across groups. Even for the same Base model, the validation loss at epoch 3 of the 10-epoch schedule (5.47) differs considerably from the final loss of the 3-epoch schedule (6.37), showing that the results of short comparison experiments are also sensitive to the learning rate schedule. Second, in the vocabulary size experiment, the parameter count and the basis for comparing loss changed together, making it difficult to isolate the effect. Third, every ex-

periment was run only once with a single fixed seed, so variation due to random initialization and data order was not evaluated. Fourth, only three model sizes were compared, so the effects of width and depth could not be separated and the optimal size could not be determined. Fifth, no downstream task evaluation was performed, so we could not verify whether differences in pre-training performance carry over to actual task performance. Sixth, as stated in Section 4.1, validation was split at the instance level and document-level independence is not guaranteed, so validation performance may have been estimated optimistically.

8. Future Work

First, we will rerun all ablations on the full data with the same number of epochs and train until the decrease in loss has sufficiently plateaued. Second, we will repeat experiments with multiple random seeds and report means and standard deviations. Third, for the vocabulary experiment, we will train all tokenizers on the same corpus, match parameter counts by adjusting the hidden dimension or by using ALBERT-style embedding factorization (Lan et al., 2020), and compare using a tokenization-independent character-level metric (e.g., bits-per-character). Fourth, we will vary width and depth in a grid to identify which aspect of capacity matters. Fifth, we will fine-tune on downstream tasks such as Korean text classification and sentence similarity to examine the relationship between pre-training performance and actual task performance.

9. Conclusion

We implemented a Korean Mini-BERT with about 1.01M parameters, pre-trained it on Korean Wikipedia, and obtained an MLM accuracy of 23.9% and an NSP accuracy of 77.4% after 10 epochs. The ablation results show that under short training, increasing model capacity lowered validation loss and raised MLM accuracy, with the largest improvement when encoder layers were added. The effects of vocabulary size and masking probability were difficult to determine due to limitations in the experimental design; confirming them would require unified training conditions, controlled parameter counts, and repeated experiments.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pp. 4171–4186, 2019.
- Kudo, T. and Richardson, J. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pp. 66–71, 2018.
- Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., and Soricut, R. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR)*, 2020.
- Park, S. et al. KLUE: Korean language understanding evaluation. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.
- Sanh, V., Debut, L., Chaumond, J., and Wolf, T. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.
- Turc, I., Chang, M.-W., Lee, K., and Toutanova, K. Well-read students learn better: On the importance of pre-training compact models. *arXiv preprint arXiv:1908.08962*, 2019.